

Noise

2 sources of Noise in a circuit:

1) External Noise

2) Internal Noise

- Ex. Resistor Noise

- Shot noise (charge crossing boundaries)

⋮

Characterizing Noise

As an example, let's assume we can define $x_n(t)$
↑ noise w.r.t time.

could get: $x_{no}(t) = A(x_n(t))$ ← A is our transfer function w.r.t time.
↑ Amplified noise output

where we would get something like:

$$X_{\odot}(t) = X_s(t) + x_{no}(t)$$

↑ my desired signal.

Better way to characterize...

use power considerations → Noise is power injected into the system.

RMS measurements of $X_s(t)$, $x_{no}(t)$

→ X_s , X_{no} ← RMS values of signal & noise.

Rms calculations:

Assume x_0 driven into $R_L = 1 \Omega$

then $p_0(t) = x_0(t)^2 / R_L = x_0(t)^2$

over period of sine wave:

~~$P_0 = \frac{1}{T} \int_0^T x_0^2(t) dt$~~ $P_0 = \frac{1}{T} \int_0^T \frac{x_0^2(t)}{R_L} dt$

$P_0 = \frac{1}{T} \int_0^T \frac{x_0^2(t)}{R_L} dt = \frac{|x_0|^2}{2 R_L}$

splitting up:

$P_0 = \underbrace{\frac{P_s}{R_L}}_{X_s^2 / R_L} + \underbrace{\frac{P_{no}}{R_L}}_{X_{no}^2 / R_L}$

X_s^2, X_{no}^2
mean squared values of
signal & noise

Signal-to-Noise Ratio ...

$SNR = 10 \log_{10} \left(\frac{X_s^2}{X_{no}^2} \right) = 20 \log_{10} \left(\frac{X_s}{X_{no}} \right)$

↑
RMS values.

for circuits:

we care about impacts on voltage & current,

Define these as

$$E_n^2 \text{ \& \; } I_n^2 \quad (\text{mean squared values})$$

$\uparrow$ voltage $\uparrow$ current.

Also define:

$$e_n(t), i_n(t)$$

$$e_n(f), i_n(f) \leftarrow \text{spectral noise densities.}$$

$$e_n^2(f), i_n^2(f) \leftarrow \text{Noise power densities}$$

$$E_n^2 = \int_{f_L}^{f_H} e_n^2(f) df, \quad I_n^2 = \int_{f_L}^{f_H} i_n^2(f) df$$

f_L, f_H defines frequency range of interest.

Total noise power:

$$E_n^2 = \int_0^{\infty} e_n^2(f) df, \quad I_n^2 = \int_0^{\infty} i_n^2(f) df.$$

f_L commonly = 0, f_H commonly $\neq \infty$

Combining Noise:

Sum via power contained: $X_{n,total}^2(f) = X_{n,1}^2(f) + X_{n,2}^2(f) + \dots$
 $X_{n,total}(f) \neq X_{n,1}(f) + X_{n,2}(f) + \dots$

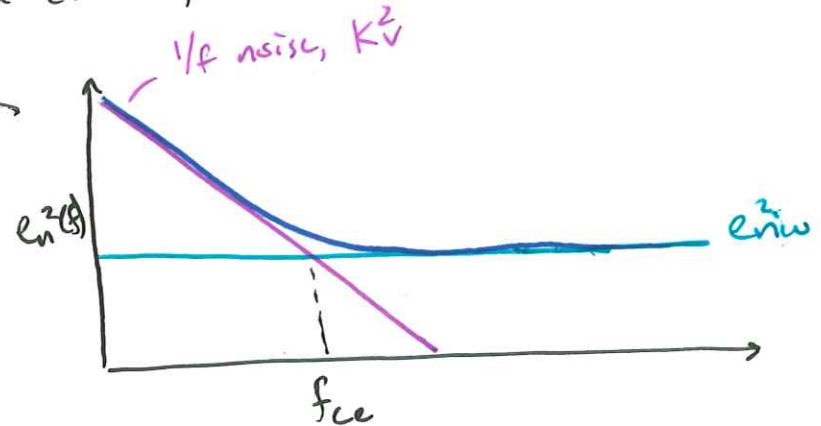
Assuming both white & $1/f$ noise in a circuit, combine!

$$e_n^2 = e_{nw}^2 + \frac{K_v^2}{f} = e_{nw}^2 \left(\frac{f_{ce}}{f} + 1 \right)$$

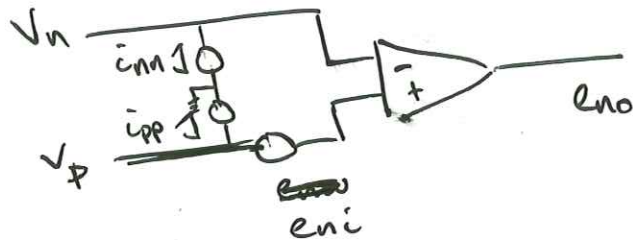
$$f_{ce} = K_v^2 / e_{nw}^2$$

$$\bar{i}_n^2 = \bar{i}_{nw}^2 + \frac{K_I^2}{f} = \bar{i}_{nw}^2 \left(\frac{f_{ci}}{f} + 1 \right)$$

$$f_{ci} = K_I^2 / \bar{i}_{nw}^2$$



As inputs to an op-amp:



white noise

$e_n(f)$, $i_n(f)$

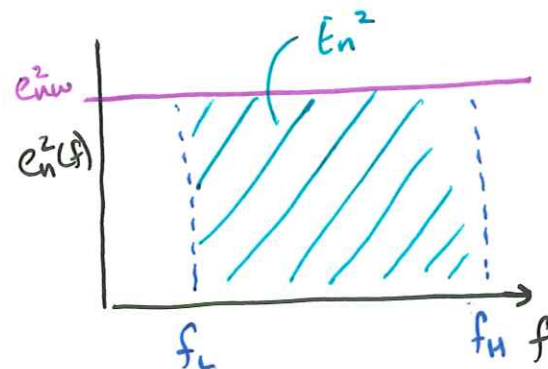
constant over frequency: $e_n(f) = e_{nw}$, $i_n(f) = i_{nw}$

$$E_n^2 = \int_{f_L}^{f_H} e_n^2(f) df = e_{nw}^2 \int_{f_L}^{f_H} df = e_{nw}^2 [f_H - f_L]$$

if $f_L \ll f_H$

$$E_n^2 \approx e_{nw}^2 f_H$$

$$I_n^2 \approx i_{nw}^2 f_H$$



1/f noise (pink noise)

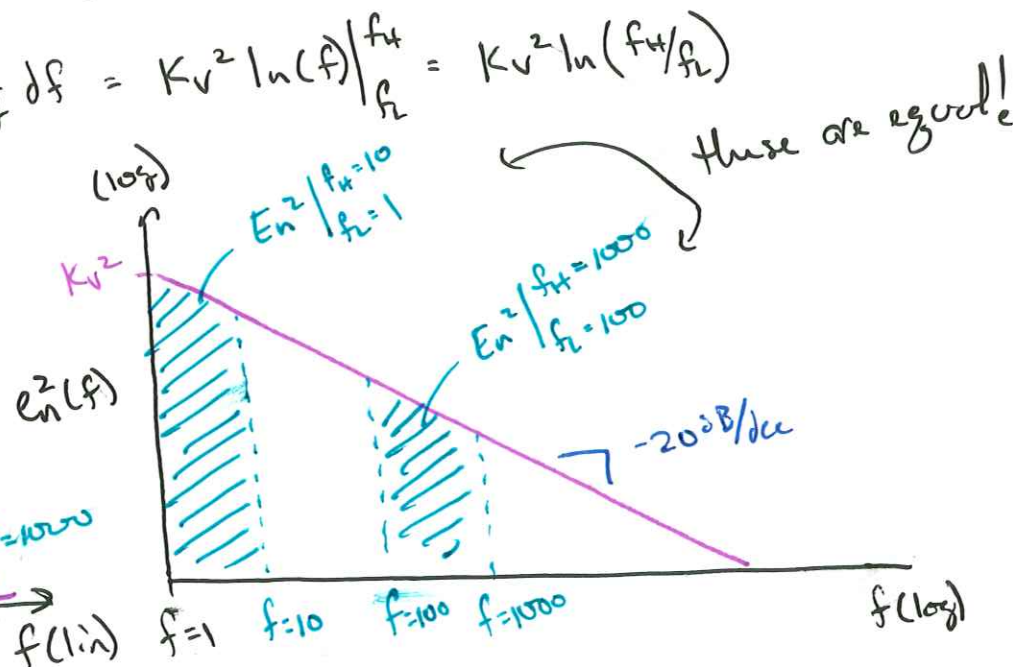
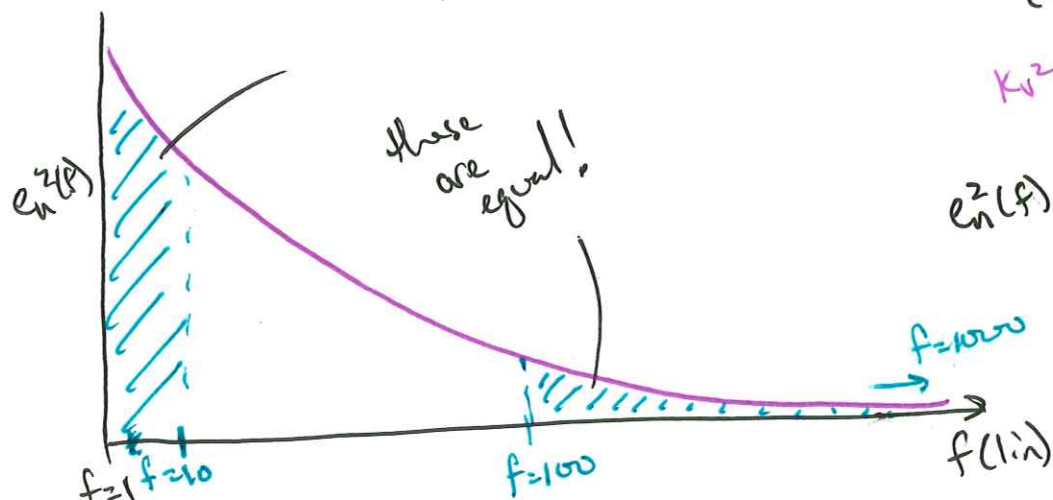
Noise increases as f decreases

$$e_n^2(f) = K_V^2 / f, \quad i_n^2(f) = K_I^2 / f$$

$$e_n(f) = K_V / \sqrt{f}, \quad i_n(f) = K_I / \sqrt{f}$$

K_V & K_I constants.

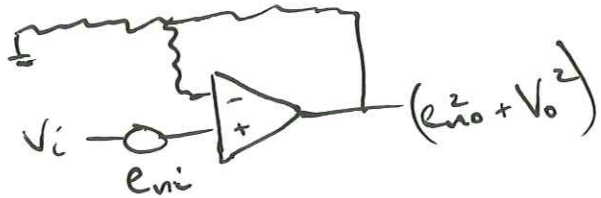
$$E_n^2 = \int_{f_L}^{f_H} e_n^2(f) df = K_V^2 \int_{f_L}^{f_H} \frac{1}{f} df = K_V^2 \ln(f) \Big|_{f_L}^{f_H} = K_V^2 \ln(f_H/f_L)$$



Let's assume $f_L \gg f_{co}, f_{ci}$ (white noise dominates) $\rightarrow e_{ni} = e_{nw}$

Amplifier with $A(jf) = A_0 / (1 + jf/f_B)$

$\rightarrow$ Add input noise



$$e_{no} = |A(jf)| e_{ni}$$

$$e_{no}^2 = |A(jf)|^2 e_{ni}^2 = |A(jf)|^2 e_{nw}^2$$

$$E_{no}^2 = \int_{f_L}^{f_H} e_{no}^2(f) df = \int_{f_L}^{f_H} |A(jf)|^2 e_{nw}^2 df = A_0^2 e_{nw}^2 \int_{f_L}^{f_H} \frac{1}{1 + f^2/f_B^2} df$$

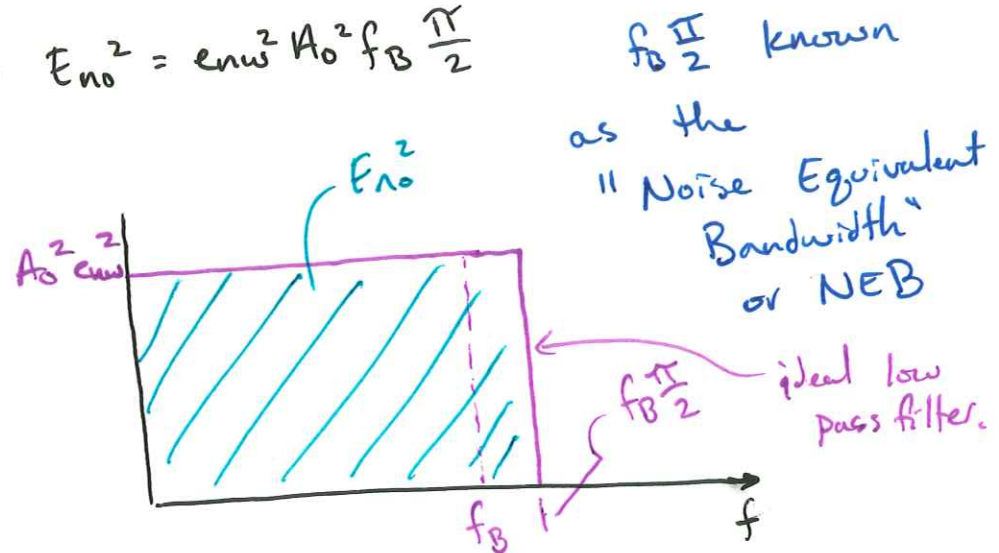
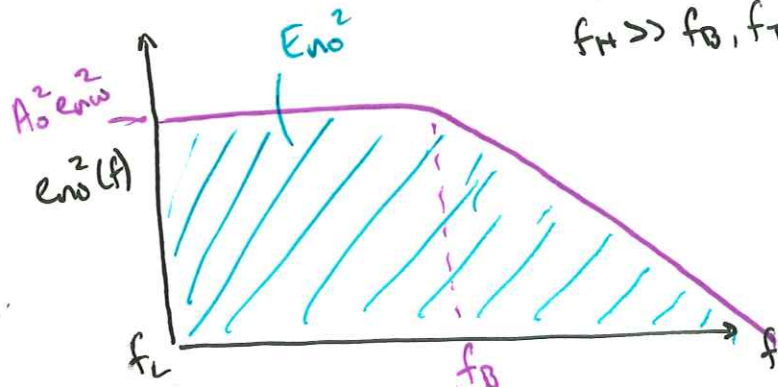
$$E_{no}^2 = A_0^2 f_B^2 e_{nw}^2 \int_{f_L}^{f_H} \frac{1}{f_B^2 + f^2} df$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} x$$

$$E_{no}^2 = A_0^2 f_B e_{nw}^2 \tan^{-1} f \Big|_{f_L}^{f_H}$$

assume $f_L \ll f_H \rightarrow E_{no}^2 = e_{nw}^2 A_0^2 f_B \frac{\pi}{2}$

$f_H \rightarrow$ very large
 $f_H \gg f_B, f_T$



Example 1

Book 7.4a

what is NEB for a cascaded (2x) amplifier, ~~the~~ both with $A(f)$
 $(f_{B1} = f_{B2})$
 Dominant-pole.



$$eno(f) = |A_1(jf)| |A_2(jf)| eni(f) = |A(jf)|^2 eni(f) \rightarrow enw$$

$$eno^2(f) = |A(jf)|^4 eni^2(f) \rightarrow enw^2$$

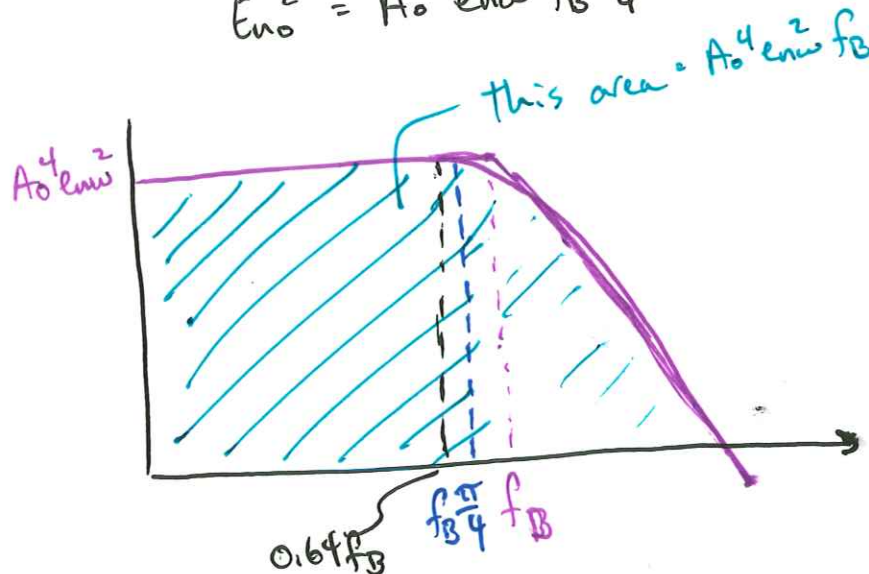
$$E_{no}^2 = \int_0^{\infty} |A(jf)|^4 enw^2 df = A_0^4 enw^2 \int_0^{\infty} \frac{1}{(1+f^2/f_B^2)^2} df$$

→ Calculator does the integration

$$E_{no}^2 = A_0^4 enw^2 f_B \frac{\pi}{4} \rightarrow NEB = f_B \frac{\pi}{4}$$

Note $\frac{\pi}{2} > 1$

$\frac{\pi}{4} < 1$



$$f_{-3dB} = \sqrt{2^{1/2} - 1} f_B = 0.64 f_B$$

$$0.64 f_B < \frac{\pi}{4} f_B$$

Example 2

Book 7.1

Characteristics of $1/f$ + white noise.

2 noise values measured: $f_1 = 10 \text{ Hz}$, $e_n(f_1) = 20 \text{ nV}/\sqrt{\text{Hz}}$

$f_2 = ?$, $e_n(f_2) = 6 \text{ nV}/\sqrt{\text{Hz}}$

know that $f_2 \gg f_{ce}$

$$e_n^2(f) = e_{nw}^2 \left(\frac{f_{ce}}{f} + 1 \right)$$

$$e_n^2(f_2) = (6 \text{ nV}/\sqrt{\text{Hz}})^2 = e_{nw}^2 \left(\frac{f_{ce}}{f_2} + 1 \right) = e_{nw}^2$$

$\therefore e_{nw} = 6 \text{ nV}/\sqrt{\text{Hz}}$

$$e_n^2(f_1) = (20 \text{ nV}/\sqrt{\text{Hz}})^2 = (6 \text{ nV}/\sqrt{\text{Hz}})^2 \left(\frac{f_{ce}}{10 \text{ Hz}} + 1 \right)$$

$$f_{ce} = 101.1 \text{ Hz} = K_V^2 / e_{nw}^2$$

$$\rightarrow K_V = 60 \text{ nV}\sqrt{\text{Hz}}$$

What is E_n over 1 mHz to 1 MHz

$$E_n^2 = \int_{1 \text{ mHz}}^{1 \text{ MHz}} \left(6 \frac{\text{nV}}{\sqrt{\text{Hz}}} \right)^2 \left(\frac{101.1 \text{ Hz}}{f} + 1 \right) df$$

$$E_n^2 = \underbrace{\left[e_{nw}^2 (f_H - f_L) \right]_{1 \text{ m}}^{1 \text{ M}}}_{\text{white noise}} + \underbrace{\left[K_V^2 \ln \left(\frac{f_H}{f_L} \right) \right]_{1 \text{ m}}^{1 \text{ M}}}_{1/f \text{ noise}} = 3.6 \times 10^{-11} \text{ V}^2$$

$\rightarrow E_n \approx 6 \text{ nV}$